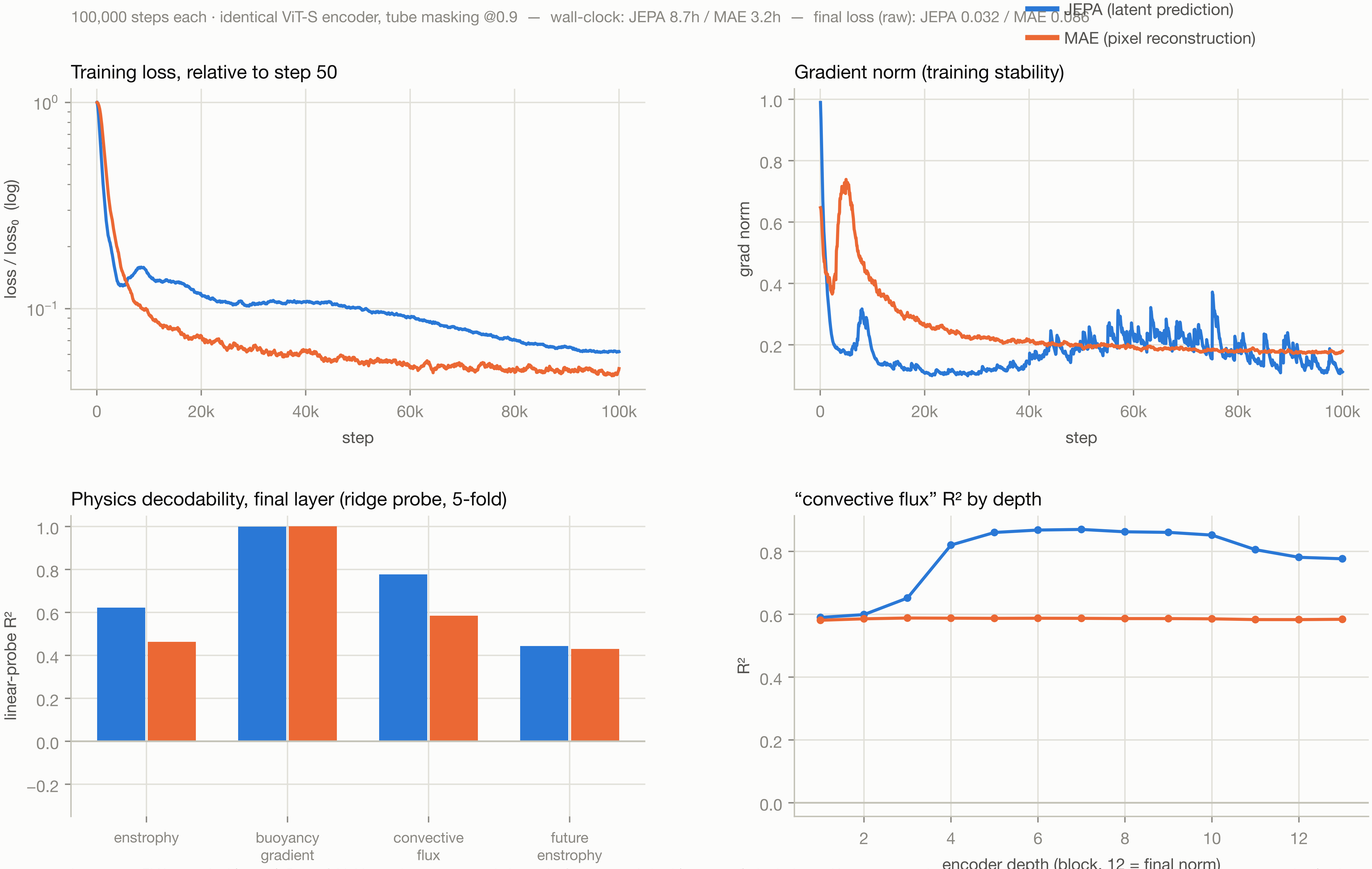
vJEPA vs vMAE — Rayleigh–Bénard

4ch · 512×128 · buoyancy-driven convection

100,000 steps each · identical ViT-S encoder, tube masking @0.9 — wall-clock: JEPA 8.7h / MAE 3.2h — final loss (raw): JEPA 0.032 / MAE 0.086



Loss curves EMA-smoothed (α=0.06). R^2 = 5-fold ridge probe on mean-pooled encoder features vs. derived (non-stored) physical quantities — not raw-channel reconstruction. Source: pod_logs/wandb offline runs + scripts/analyze_encoder